Feature importance of the 15:30 forecast (16:00 bar) re-run on the de-duplicated per-bar design

MDI, split count, permutation importance on the causal held-out tail, SHAP — trees and the per-bar linear models; 1,469 forecasts 2018-06-25 .. 2024-04-30, 147 refits

What changed from the first pass. (i) The design: the per-bar design no longer carries the twelve HAR×open/close session-edge interactions (at 16:00 the six `har_ma_k_x_open` columns were all zero and the six `har_ma_k_x_close` columns exact copies of `har_ma_1 .. har_ma_3125`); columns HAR + calendar 34→22, all features 640→628. (ii) The trees: every tree fit uses only the columns kept by the per-window mask of its own 2000-session training window (constant columns and exact copies removed; the linear arms' identifiability mask applied to the trees); MDI, split count and TreeSHAP are mapped back to all columns with 0 for a dropped column, and permuting a dropped column changes nothing. Kept columns per refit (min / median / max): HAR + calendar 17 / 17 / 17 of 22; all features 360 / 369 / 378 of 628. The linear models are unchanged by the mask (theirs already removed these columns at every tune). Everything else is the first pass's (147 refits, the same held-out tails, permutation generator and seeds, models and parameters), so before/after differs only by the design and the mask — and by the permutation draws of most units (the stream is laid out unit by unit and the unit count changed: a column's P1 draws are identical in both runs, the P2 draws and the group units' draws are fresh; the ridge fits are the same problem in both runs, so the ridge rows measure that Monte Carlo noise). (iii) Cadence: the campaign's trees now refit every session; at an importance refit the model is identical under either cadence (same window, same configuration), so the importance at every 10th refit is the daily-refit model's importance on those days (Section 8). Before/after rankings: Section 9.

Setup. Models: per-bar ridge and lasso (coefficients re-solved every session, penalty re-chosen causally every 250 sessions) and the untuned per-bar LightGBM, XGBoost and random forest (refit every 10 sessions) — the shipped configurations. Every fit uses the 2000 sessions before it. A refit's *held-out tail* is the $\leq$ 10 sessions its model forecasts, up to the next refit: never in its fit, so a loss change measured there is out of sample and causal. Buckets: all features, and HAR + calendar. Library: scikit-learn 1.9.0 (random forest), LightGBM 4.6.0, XGBoost 3.2.0, shap 0.51.0; the permutation and drop-column loops are written out, because a library permutation routine scores one fitted model on one test set, not 147 walk-forward models on their own tails.

Measures. (1) *MDI / gain*: LightGBM gain, XGBoost total gain, random-forest impurity decrease; share per refit, averaged. (2) *Split count.* (3) *Permutation importance* on the tail, loss = QLIKE of the plain back-transform $\hat{y}^2\times$ diurnal baseline against the bar's realized variance (no recalibration), and squared error in the fit space; 10 draws per unit and refit, the same draws for all five models. P1 shuffles the unit's values among the tail's rows (the literal permutation; blind to slow inputs, which barely move in 10 sessions); P2 replaces them with rows drawn from the refit's own training window (past data only). A unit is a column, a *series* (all columns of one input: lags, flags, gates) or a *cluster* (columns every pair of which has $|\text{corr}| \geq 0.8$, complete linkage); a group's columns move together. Intervals: 95% circular-block bootstrap over refits (block 6). (4) *SHAP*: TreeSHAP (path-dependent) on the same tail rows; for the linear models $\beta_j(x_j - \bar{x}_j)$ with the window mean as background. Linear extras: $|\beta \cdot sd|$ and drop-column importance (re-solve without the unit, penalty held).

Key. Every exogenous input is divided by its time-of-day baseline, and the HAR columns `har_ma_*` are trailing means of the adjusted target, so no name carries a prefix. `har_ma_*` realized variance of the bar (the target's own HAR ladder, means over the last 1, 5, 25, 125, 625, 3125 bars; the per-bar design no longer carries the session-edge interactions `har_ma_*_x_open/close`); `sumabsret_ma_*` absolute return; `sumret_ma_*` signed return; `sumret3/4_ma_*` 3rd/4th-power returns; `sumpret2_ma_*` upside squared returns; `sumbipow_ma_*` bipower variation; `sumautocov_ma_*` return autocovariance; `sumvolume_ma_*` ES volume; `numobs_ma_*` ES prints per bar; `vix/vvix/vix3m_ma_*` the Cboe indices; `fomc_*` FOMC flags and distances; `*_avail/active_ma_*` is-present / is-nonzero flags; calendar = `DOW_*`, `is_*`, `hour`, `days_to_opex`. all features adds the constituent cross-section (`*_ewstock`, `*_vwstock`), Cboe volume (`voldemand_*`) and StockTwits (`stocktwits_*`).

Table 1: The design is float throughout, but not all continuous: distinct values per column in each 2000-session window (median over the 147 windows) — what MDI's cardinality bias keys on.

bucket	columns	constant	binary	3–25	26–500	>500	series	clusters (columns)
HAR + calendar	22	4	11	0	1	6	2	3 (6)
all features	628	137	40	93	91	267	49	100 (498)

1. Does MDI's cardinality bias matter here? Yes, below the leader.

Table 2: Spearman rank correlation, over the non-constant columns, between a column's importance and its number of distinct values. MDI, split count and TreeSHAP track cardinality; permutation importance does not.

bucket	model	MDI	split	SHAP	perm P1 QL	perm P2 QL	perm P2 MSE
all features	LightGBM	0.82	0.82	0.79	-0.02	-0.04	-0.05
all features	XGBoost	0.82	0.82	0.80	0.18	0.17	0.03
all features	random forest	0.82	0.83	0.81	0.00	-0.06	-0.13
HAR + calendar	LightGBM	0.86	0.86	0.81	0.37	0.41	0.41
HAR + calendar	XGBoost	0.84	0.86	0.79	0.37	0.55	0.56
HAR + calendar	random forest	0.86	0.86	0.81	0.42	0.56	0.79

Table 3: Noise probes: on every third refit a second fit appends three pure-noise columns (continuous; 20 values; binary). Median rank among all columns (1 = most important) and, in brackets, the share of the real columns the model uses that rank below the probe; permutation P2 of the continuous probe (ΔQLIKE $\times 10^3$, 95% interval) against `har_ma_*`'s.

bucket	model	cols	MDI rank	split rank	SHAP rank	20-val MDI	binary MDI	perm P2 (cont.)	`har_ma_*`
all features	LightGBM	631	73 (77%)	47 (85%)	82 (70%)	138	315	-0.07 [-0.39, $+0.24$]	249
all features	XGBoost	631	109 (61%)	75 (69%)	151 (54%)	169	309	$+0.04$ [-0.06, $+0.17$]	236
all features	random forest	631	45 (86%)	11 (97%)	81 (71%)	109	310	-0.02 [-0.38, $+0.30$]	361
HAR + calendar	LightGBM	25	7 (68%)	7 (68%)	8 (59%)	7	10	$+0.32$ [-1.62, $+2.67$]	636
HAR + calendar	XGBoost	25	6 (69%)	6 (72%)	9 (55%)	8	13	$+0.59$ [-0.28, $+1.54$]	587
HAR + calendar	random forest	25	5 (73%)	3 (88%)	7 (64%)	7	9	-0.24 [-2.13, $+1.78$]	690

Reading. A column of pure noise with a new value on every row is ranked by MDI above 77%, 61%, 86% (LightGBM, XGBoost, forest) of the real all-features columns the model uses, and by the forest's split count above 97%; the binary noise column sits near the bottom. The same noise gets a permutation importance of at most 0.59×10^{-3} in absolute value (6 of 6 intervals include zero), against at least 0.24 for `har_ma_*`. TreeSHAP, which splits the fitted trees, inherits part of the bias. Over the real columns, MDI, split count and SHAP correlate 0.82–0.81 with the number of distinct values, permutation -0.06–0.18. So below the leader, the MDI / split ranking of this design is partly a ranking by cardinality; the permutation ranking is the one to use for selection. At the series level: `har_ma_*` is the leader of 35 of 36 tree bucket x model x measure rows; not of all features random forest split (leader `voldemand_spx_open_and_close`). The cardinality bias matters for everything below the leader. Series that MDI or split count puts in the top 5 but permutation P2 ranks outside the top 10 (permutation rank in brackets): all features LightGBM: `sellturnover_vwstock_ma_*` (36), `stocktwits_sentiment_ma_*` (49), `sumret3_ma_*` (42); all features XGBoost: none; all features random forest: `voldemand_spx_open_and_close_ma_*` (34), `sumret3_ewstock_ma_*` (29), `voldemand_all_open_only_ma_*` (39).

2. The top inputs by measure

Table 4: All-features design, series level: top three per model and measure (share for MDI / split / $|\beta \cdot sd|$ / SHAP; change as % of the model's tail loss for permutation and drop-column).

model	measure	top three
ridge	$\|\beta \cdot sd\|$	har_ma_* 11%; sumabsret_ma_* 5%; sumabsret_vwstock_ma_* 5%
ridge	perm. P1, ΔMSE	har_ma_* +69%; sumabsret_vwstock_ma_* +8%; sumabsret_ma_* +7%
ridge	perm. P2, ΔMSE	har_ma_* +142%; sumabsret_ma_* +22%; sumret4_ma_* +18%
ridge	perm. P2, ΔQLIKE	voldemand_spx_open_only_ma_* +59024%; stocktwits_sentiment_ma_* +4255%; sumret3_ma_* +3480%
ridge	drop-column, ΔMSE	har_ma_* +17%; calendar +2%; sumvolume_ma_* +2%
ridge	SHAP mean $\|\phi\|$	har_ma_* 15%; sumabsret_ma_* 6%; sumabsret_vwstock_ma_* 5%
lasso	$\|\beta \cdot sd\|$	har_ma_* 46%; sumabsret_vwstock_ma_* 10%; sumret_ma_* 9%
lasso	perm. P1, ΔMSE	har_ma_* +156%; sumabsret_vwstock_ma_* +9%; sumret_ma_* +7%
lasso	perm. P2, ΔMSE	har_ma_* +294%; sumabsret_vwstock_ma_* +17%; vix_ma_* +15%
lasso	perm. P2, ΔQLIKE	effspread_vwstock_ma_* +569%; har_ma_* +259%; sumpret2_vwstock_ma_* +73%
lasso	drop-column, ΔMSE	har_ma_* +27%; sumret_ma_* +2%; sumabsret_vwstock_ma_* +1%
lasso	SHAP mean $\|\phi\|$	har_ma_* 46%; sumabsret_vwstock_ma_* 10%; sumret_ma_* 9%
LightGBM	MDI / gain	har_ma_* 52%; sumabsret_ma_* 6%; sumret_ma_* 5%
LightGBM	split count	har_ma_* 10%; sumret_ma_* 6%; sumvolume_ma_* 4%
LightGBM	perm. P1, ΔQLIKE	har_ma_* +97%; calendar +5%; sumret_ma_* +4%
LightGBM	perm. P2, ΔQLIKE	har_ma_* +200%; calendar +6%; sumret_ma_* +3%
LightGBM	perm. P2, ΔMSE	har_ma_* +204%; calendar +4%; sumvolume_ma_* +3%
LightGBM	SHAP mean $\|\phi\|$	har_ma_* 44%; sumret_ma_* 6%; sumvolume_ma_* 5%
XGBoost	MDI / gain	har_ma_* 64%; sumabsret_ma_* 7%; sumret_ma_* 4%
XGBoost	split count	har_ma_* 19%; sumret_ma_* 8%; sumvolume_ma_* 5%
XGBoost	perm. P1, ΔQLIKE	har_ma_* +95%; calendar +4%; sumret_ma_* +3%
XGBoost	perm. P2, ΔQLIKE	har_ma_* +187%; calendar +5%; sumvolume_ma_* +5%
XGBoost	perm. P2, ΔMSE	har_ma_* +193%; sumvolume_ma_* +4%; calendar +3%
XGBoost	SHAP mean $\|\phi\|$	har_ma_* 51%; sumret_ma_* 7%; sumvolume_ma_* 6%
random forest	MDI / gain	har_ma_* 62%; sumabsret_ma_* 5%; sumret_ma_* 3%
random forest	split count	voldemand_spx_open_and_close_ma_* 3%; har_ma_* 3%; voldemand_all_open_and_close_ma_* 3%
random forest	perm. P1, ΔQLIKE	har_ma_* +134%; calendar +3%; sumabsret_ma_* +2%
random forest	perm. P2, ΔQLIKE	har_ma_* +283%; calendar +4%; sumabsret_ma_* +3%
random forest	perm. P2, ΔMSE	har_ma_* +246%; calendar +2%; sumvolume_ma_* +1%
random forest	SHAP mean $\|\phi\|$	har_ma_* 58%; sumabsret_ma_* 5%; sumret_ma_* 4%

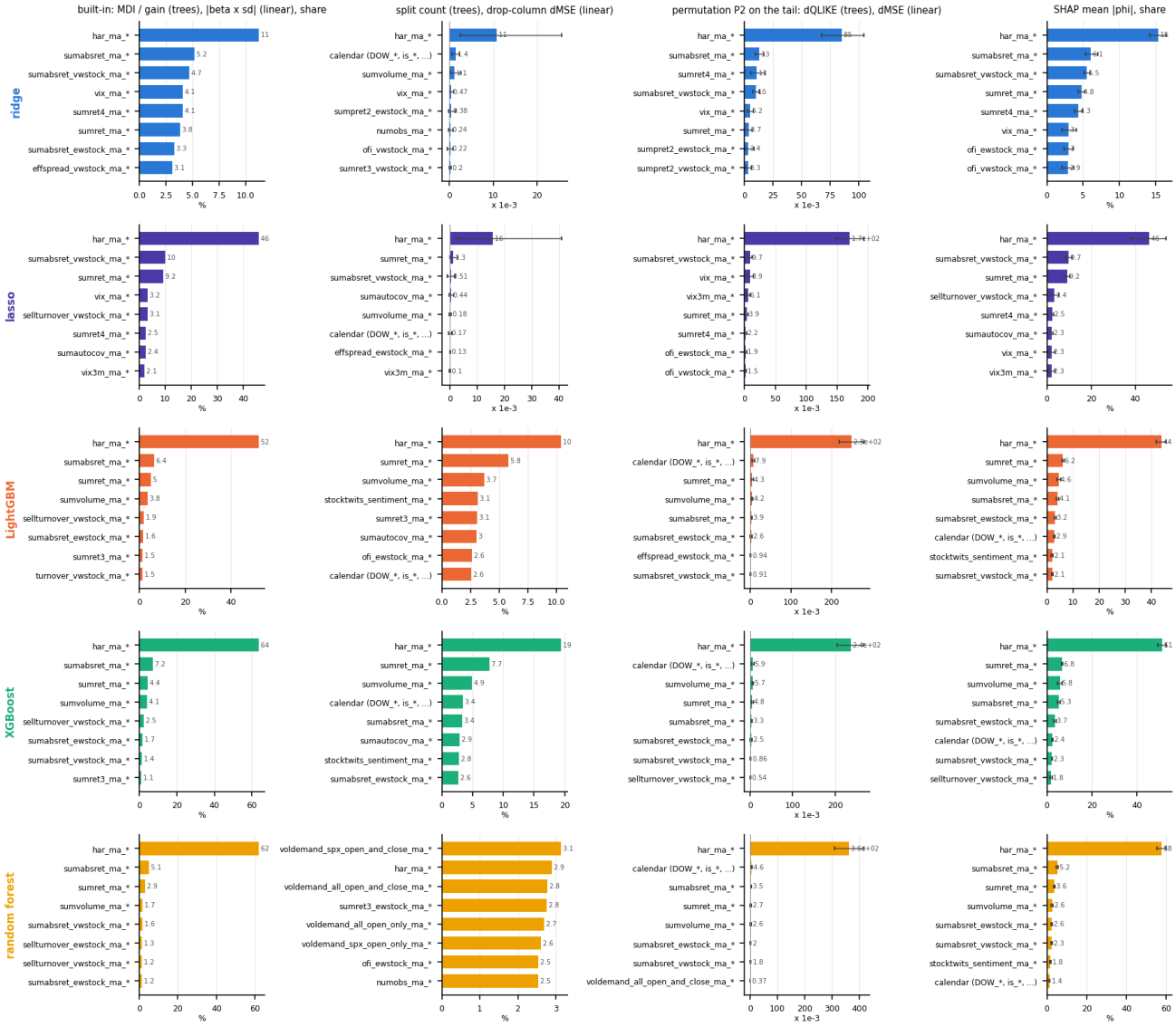


Figure 1: All features: ranked bars per measure (series level; top 8). Rows: models; columns: the built-in measure, split count (trees) / drop-column (linear), permutation P2 on the tail, SHAP. Bars with 95% intervals over refits. Linear permutation / drop-column in ΔMSE (see Section 6).

3. Ranking agreement

Table 5: Within a tree model: Spearman between measures (series / column level).

bucket	level	model	MDI∼split	MDI∼SHAP	MDI∼P2	SHAP∼P2	P1∼P2	P2 QL∼MSE
all features	series	LightGBM	0.87	0.87	0.21	0.27	0.69	0.72
all features	series	XGBoost	0.87	0.88	0.52	0.56	0.78	0.58
all features	series	random forest	0.55	0.86	0.17	0.31	0.68	0.68
all features	column	LightGBM	0.99	0.97	0.06	0.05	0.53	0.60
all features	column	XGBoost	0.99	0.97	0.26	0.25	0.55	0.47
all features	column	random forest	0.98	0.98	0.03	0.03	0.45	0.51

Table 6: Within a linear model: Spearman between measures.

bucket	level	model	$\lvert\beta sd\rvert$ ∼SHAP	$\lvert\beta sd\rvert$ ∼P2	SHAP∼P2	P1∼P2	P2∼drop	P2 QL∼MSE
all features	series	ridge	0.83	0.59	0.73	0.77	0.38	-0.05
all features	series	lasso	0.95	0.88	0.86	0.82	0.13	0.95
all features	column	ridge	0.99	0.42	0.42	0.58	0.16	0.09
all features	column	lasso	0.99	0.40	0.38	0.65	0.13	0.78

Table 7: Across models, same measure: range of Spearman over tree–tree pairs, tree–linear pairs, and ridge–lasso.

bucket	level	SHAP			perm P2 ΔQLIKE			perm P2 ΔMSE		
		tree	tree–lin	r–l	tree	tree–lin	r–l	tree	tree–lin	r–l
all features	series	0.83–0.89	0.32–0.49	0.86	0.39–0.61	-0.26–0.37	-0.10	0.49–0.58	0.01–0.49	0.69
all features	column	0.95–0.97	0.68–0.88	0.78	0.17–0.27	-0.13–0.10	0.09	0.23–0.26	-0.00–0.07	0.43

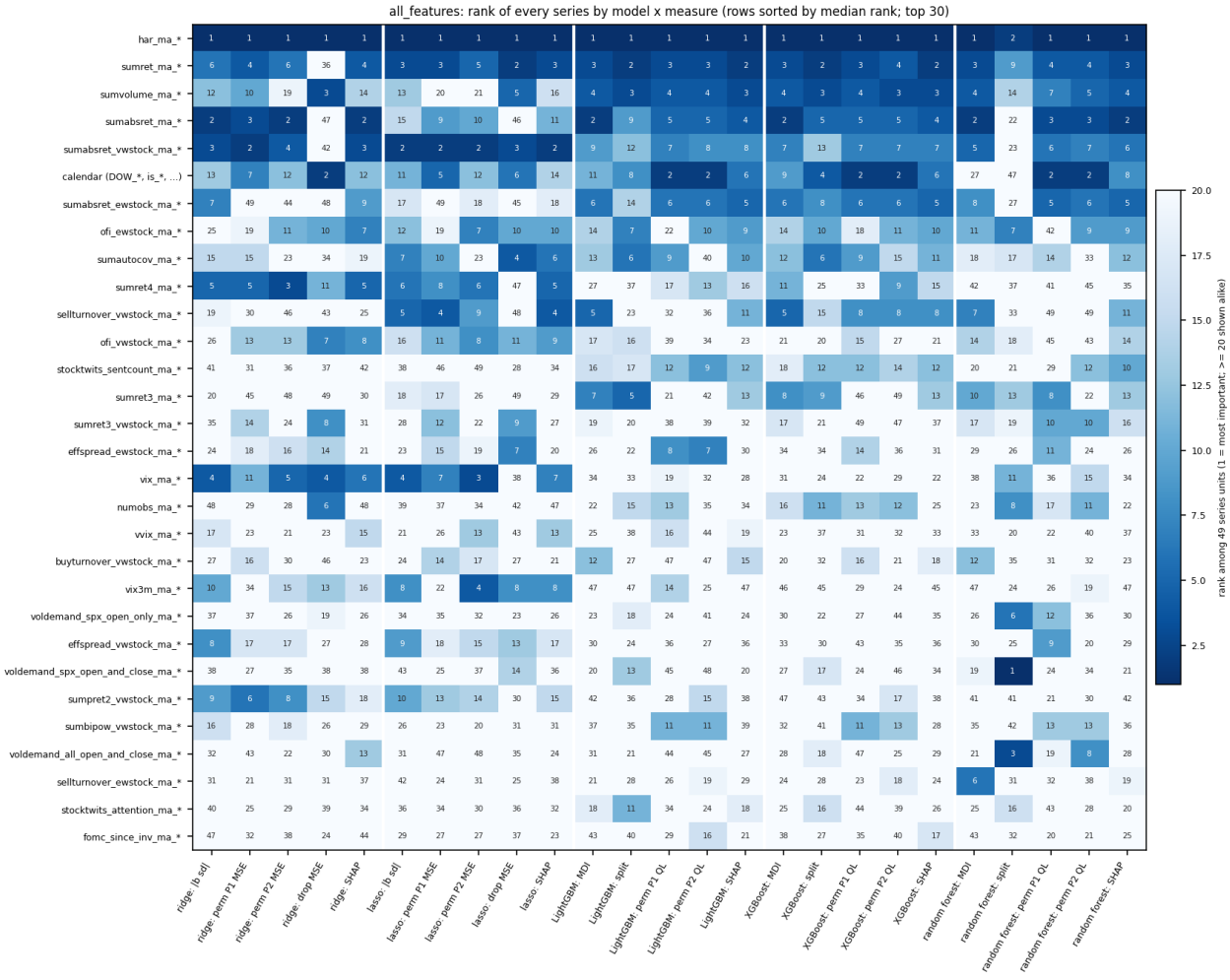


Figure 2: All features: rank of every series under every model × measure (1 = most important).

4. Stability over refits and regimes

Table 8: Top-5 stability (series level): share of the 147 refits in which `har_ma_*` is in the refit's top 5; number of series in the top 5 in at least half the refits; number ever in the top 5. Measures: built-in (MDI or $|\beta sd|$), permutation P2 (ΔQLIKE trees, ΔMSE linear), SHAP.

bucket	model	built-in			perm P2			SHAP		
		har	≥50%	ever	har	≥50%	ever	har	≥50%	ever
all features	ridge	100%	5	11	98%	3	44	99%	3	27
all features	lasso	100%	4	12	100%	3	30	100%	3	24
all features	LightGBM	100%	5	10	99%	2	47	100%	4	28
all features	XGBoost	100%	5	10	99%	3	47	100%	4	15
all features	random forest	100%	4	13	99%	1	48	100%	5	29

Table 9: Regimes: Spearman between the ranking within a stratum and the full-sample ranking (series level): median (minimum) over the calendar years (2018: 13, 2019: 26, 2020: 25, 2021: 25, 2022: 25, 2023: 25, 2024: 8 refits) and over the causal VIX terciles (VIX at 15:30 against the 1/3, 2/3 quantiles of the previous 2000 sessions' 15:30 VIX: low 28, mid 38, high 81 refits); years in which the full-sample leader also leads.

bucket	model	built-in			perm P2			SHAP		
		year	VIX	lead	year	VIX	lead	year	VIX	lead
all features	ridge	0.93 (0.84)	0.99 (0.95)	7/7	0.67 (0.50)	0.78 (0.60)	7/7	0.76 (0.54)	0.92 (0.83)	7/7
all features	lasso	0.60 (0.47)	0.99 (0.84)	7/7	0.39 (0.15)	0.72 (0.31)	7/7	0.64 (0.42)	0.97 (0.83)	7/7
all features	LightGBM	0.94 (0.87)	0.99 (0.98)	7/7	0.60 (0.31)	0.66 (0.61)	7/7	0.83 (0.73)	0.93 (0.93)	7/7
all features	XGBoost	0.94 (0.88)	0.99 (0.98)	7/7	0.57 (0.10)	0.65 (0.51)	7/7	0.87 (0.66)	0.93 (0.92)	7/7
all features	random forest	0.98 (0.94)	0.99 (0.98)	7/7	0.49 (0.32)	0.60 (0.31)	7/7	0.90 (0.77)	0.97 (0.96)	7/7

5. Correlated inputs: the dilution caveat

Table 10: Joint permutation of a cluster against the sum of its columns permuted one at a time (P2, ΔMSE in the fit space $\times 10^3$, joint / sum; squared error so the linear columns are not dominated by single tails), for the four clusters with the largest joint effect in LightGBM. Joint > sum: the columns substitute for one another and one-at-a-time permutation understates the group (dilution). Joint < sum: offsetting weights (permuting one column breaks an offset its partner carries).

bucket	cluster	members	ridge	lasso	LightGBM	XGBoost	forest
all features	C49	`har_ma_1`, `har_ma_5` (2; $\|r\|$ ≥0.89)	33.2 / 18.3	85.7 / 50.0	86.9 / 45.6	80.2 / 45.3	139.1 / 75.9
all features	C46	`har_ma_25`, `har_ma_125` (2; $\|r\|$ ≥0.83)	13.3 / 7.3	20.0 / 11.9	6.4 / 4.2	7.0 / 4.6	3.1 / 2.2
all features	C51	`sumabsret_ma_1`, `sumabsret_ma_5`, `sumret4_ma_1`, `sumret4_ma_5` +4 (8; $\|r\|$ ≥0.82)	7.2 / 3.6	0.5 / 2.6	1.7 / -0.6	2.1 / -0.8	0.7 / -0.8
all features	C50	`sumabsret_ewstock_ma_1`, `sumabsret_ewstock_ma_5`, `sumabsret_vwstock_ma_1`, `sumabsret_vwstock_ma_5` (4; $\|r\|$ ≥0.85)	15.5 / 3.9	8.7 / 7.0	1.4 / 0.6	0.3 / -0.4	1.4 / 0.4

6. Linear models: QLIKE permutation is dominated by single tails

The linear forecasts are unbounded: moving an extreme input (a crash day's 4th-power or absolute return) onto another row can push the fit-space forecast through zero, and QLIKE of $\hat{y}^2\times$ baseline explodes. Every permutation / drop-column table therefore carries the median refit and the largest single-refit share of the sum, and the linear figures and cross-model comparisons use the squared-error versions. Trees cannot extrapolate beyond the training range of the target, so their QLIKE changes are well behaved.

Table 11: The top linear QLIKE entries: mean [95% interval], median refit, share of the sum carried by one refit.

bucket	model	measure	top series	mean [interval]	median	one refit
all features	ridge	perm. P1, ΔQLIKE	`sumautocov_ma_*`	492 [0.00069, 1.5e+03]	0.0011	100%
all features	ridge	perm. P2, ΔQLIKE	`voldemand_spx_open_only_ma_*`	1.5e+03 [0.0011, 4.5e+03]	0.0011	100%
all features	ridge	drop-column, ΔQLIKE	`sumabsret_ma_*`	980 [0.0016, 2.9e+03]	-0.00025	100%
all features	lasso	perm. P1, ΔQLIKE	`har_ma_*`	0.148 [0.12, 0.17]	0.12	5%
all features	lasso	perm. P2, ΔQLIKE	`effspread_vwstock_ma_*`	0.706 [0.0025, 2]	5.8e-05	89%
all features	lasso	drop-column, ΔQLIKE	`har_ma_*`	38.7 [0.0065, 1.2e+02]	0.0052	100%

7. Tuned trees

Table 12: Causally tuned trees (their own extracts: gain per refit and TreeSHAP rows at 16:00): `har_ma_*` share tuned (untuned), rank correlation tuned vs untuned over the series, and the next three series by tuned SHAP. Split count and permutation were not computed for the tuned trees (their per-refit configurations would have to be refitted).

bucket	model	MDI har	SHAP har	ρ MDI	ρ SHAP	next by tuned SHAP
all features	LightGBM	51% (52%)	40% (44%)	0.90	0.90	`sumabsret`, `sumvolume`, `sumret`
all features	XGBoost	50% (64%)	41% (51%)	0.95	0.91	`sumvolume`, `sumabsret`, `sumret`
all features	random forest	44% (62%)	41% (58%)	0.62	0.63	`sumabsret`, `sumvolume`, `sumabsret_vwstock`

8. Gates

- Design (`gates_design.csv`): baseline p 22 (want 22; old 34), no `_x_` column: True, old minus new = the 12 session-edge columns: True; all_features p 628 (want 628; old 640), no `_x_` column: True, old minus new = the 12 session-edge columns: True; the same out-of-sample stamps, the same targets and every other column bit for bit as the first pass's design: True; the old x_open columns all zero and the old x_close columns = `har_ma_k` bit for bit: True. Stamps, targets and column names also equal the stored de-dup tree runs'.
- Refits vs the stored forecasts (the reproduction gate, no-mask root vs the stored de-dup T10 runs, unmasked, agent B): bit for bit ($\leq$ 1e-9) in 5 of 6 bucket x model arms; not in all features LightGBM (max |gap| 0.054, mean 0.0095, correlation 0.9995, 147 of 147 refits above 1e-9). The LightGBM gap is the first pass's again, now with ONE thread on both sides (the thread count is ruled out), and forcing LightGBM's column-wise or row-wise histograms leaves it unchanged; the refit is deterministic run to run and the cluster and a laptop give the same gap, so the stored LightGBM runs differ in something the input file does not carry. The LightGBM importance is that of the refit.
- The mask's effect on the forecasts (masked refits vs the same unmasked stored runs): max |gap| HAR + calendar LightGBM 0.138, HAR + calendar XGBoost 0.199, HAR + calendar random forest 0.234, all features LightGBM 0.135, all features XGBoost 0.172, all features random forest 0.338; correlation 0.9974..0.9993.
- Masked refits vs the canonical masked tree tables (`results/spxw_pnl/yhat_subtree_<model>_<bucket>`, agent H): bit for bit ($\leq$ 1e-9) in 4 of 6 arms; otherwise max |gap| all features LightGBM 0.0503, all features random forest 0.00228 (`gates_canonical.csv`).
- Cadence identity – nomask (6 bucket x model arms, 147 importance refits each): every-session vs every-10 run on the importance-refit days, forecast max |gap| 0.0e+00, native importance max |gap| 0.0e+00; this study's refit vs the every-session run: forecast equal ($\leq$ 1e-9) in 5 of 6 arms (not in all features LightGBM: 4.1e-02); mask (6 bucket x model arms, 147 importance refits each): every-session vs every-10 run on the importance-refit days, forecast max |gap| 0.0e+00, native importance max |gap| 0.0e+00; this study's refit vs the every-session run: forecast equal ($\leq$ 1e-9) in 4 of 6 arms (not in all features LightGBM: 5.0e-02, all features random forest: 2.3e-03).
- The linear coefficient captures (re-run on the new design through the spec's own class): coefficient x row + intercept = the forecast; capture vs the stored de-dup research forecasts (agent A, `results/linear_subsection_dedup/arms_hof` ridge max rel HAR + calendar 6e-11, all features 4e-11; lasso HAR + calendar 6.5e-15, all features 1.2e-03 (71 of 1469 rows above 1e-9). The lasso gaps are the warm-homotopy float path of the spec's lasso: it moves with the CPU architecture (agent A: a re-run on the same architecture is bit-identical, another machine moves 16:00 forecasts by 5e-4..8e-2), and the stored arms ran on the other cluster; the lasso importance is that of this capture. Drop-column anchors reproduce the captures: ridge $\leq$ 2e-10, lasso $\leq$ 2e-02 (above 1e-6 on 0 / 9 of 147 refits, HAR + calendar / all features). The all_features lasso refits 75..124 (penalty 0.001) ran on the cluster (`cluster/slurm/submit_featimp_dedup.sh linear`), the rest locally; the parts partition the 147 refits (checked by the merge).
- Every model of a bucket saw the identical permutation draws (md5 per refit equal across the five models): True. TreeSHAP additivity: sum of phi + expected value = forecast to $\leq$ 3e-06.

9. Before / after: first pass (old design) vs de-dup + mask

Every unit of every measure and level, old vs new: `before_after_ranks_<bucket>_<model>.csv`; clusters are renumbered per run and matched by their members once the session-edge columns are removed from the old cluster. Spearman between the old and the new importance over the units common to both runs:

Table 13: Spearman, old vs new importance over the common units: median (minimum) over the model's measures. Trees also old vs de-dup without the mask and no mask vs mask.

bucket	level	trees old–new	linear old–new	trees old–no mask	trees no mask–mask
HAR + calendar	series	–	–	–	–
HAR + calendar	cluster	0.99 (0.88)	1.00 (0.96)	0.98 (0.84)	0.99 (0.86)
HAR + calendar	column	0.97 (0.75)	1.00 (0.98)	0.98 (0.81)	0.98 (0.90)
all features	series	0.86 (0.68)	0.99 (0.87)	0.87 (0.57)	0.88 (0.66)
all features	cluster	0.79 (0.59)	0.99 (0.77)	0.79 (0.56)	0.80 (0.62)
all features	column	0.63 (0.38)	1.00 (0.79)	0.60 (0.42)	0.60 (0.40)

Table 14: all features: `har_ma_*` (series) old / de-dup no mask / new: share for MDI, split, $|\beta sd|$, SHAP; permutation and drop-column as % of the model's tail loss. *x_close share*: the old `_x_close` copies' part of the old `har_ma` column total.

model	measure	old	no mask	new	x_close share	rank old	rank new
ridge	abs(b sd)	11%		11%	0%	1	1
ridge	SHAP	15%		15%	0%	1	1
ridge	perm P2 MSE	+142%		+142%	0%	1	1
ridge	drop MSE	+17%		+17%	0%	1	1
lasso	abs(b sd)	46%		46%	0%	1	1
lasso	SHAP	46%		46%	0%	1	1
lasso	perm P2 MSE	+290%		+294%	0%	1	1
lasso	drop MSE	+27%		+27%	0%	1	1
LightGBM	MDI	57%	52%	52%	22%	1	1
LightGBM	split	12%	10%	10%	22%	1	1
LightGBM	SHAP	47%	44%	44%	22%	1	1
LightGBM	perm P2 QL	+237%	+198%	+200%	9%	1	1
LightGBM	perm P2 MSE	+235%	+203%	+204%	4%	1	1
XGBoost	MDI	69%	66%	64%	22%	1	1
XGBoost	split	21%	20%	19%	24%	1	1
XGBoost	SHAP	55%	53%	51%	24%	1	1
XGBoost	perm P2 QL	+224%	+203%	+187%	13%	1	1
XGBoost	perm P2 MSE	+231%	+210%	+193%	10%	1	1
random forest	MDI	62%	62%	62%	54%	1	1
random forest	split	4%	3%	3%	50%	1	2
random forest	SHAP	58%	58%	58%	53%	1	1
random forest	perm P2 QL	+290%	+286%	+283%	54%	1	1
random forest	perm P2 MSE	+252%	+246%	+246%	58%	1	1

Table 15: HAR + calendar: `har_ma_*` (series) old / de-dup no mask / new: share for MDI, split, $|\beta sd|$, SHAP; permutation and drop-column as % of the model's tail loss. *x_close share*: the old `_x_close` copies' part of the old `har_ma` column total.

model	measure	old	no mask	new	x_close share	rank old	rank new
ridge	abs(b sd)	81%		81%	0%	1	1
ridge	SHAP	90%		90%	0%	1	1
ridge	perm P2 MSE	+501%		+501%	0%	1	1
ridge	drop MSE	+247%		+247%	0%	1	1
lasso	abs(b sd)	84%		84%	0%	1	1
lasso	SHAP	91%		91%	0%	1	1
lasso	perm P2 MSE	+495%		+495%	0%	1	1
lasso	drop MSE	+244%		+244%	0%	1	1
LightGBM	MDI	96%	95%	95%	24%	1	1
LightGBM	split	87%	84%	85%	23%	1	1
LightGBM	SHAP	91%	90%	90%	23%	1	1
LightGBM	perm P2 QL	+483%	+490%	+488%	8%	1	1
LightGBM	perm P2 MSE	+426%	+432%	+430%	2%	1	1
XGBoost	MDI	97%	97%	97%	30%	1	1
XGBoost	split	82%	79%	79%	26%	1	1
XGBoost	SHAP	93%	93%	93%	28%	1	1
XGBoost	perm P2 QL	+457%	+457%	+457%	18%	1	1
XGBoost	perm P2 MSE	+444%	+442%	+444%	14%	1	1
random forest	MDI	95%	95%	95%	46%	1	1
random forest	split	82%	76%	76%	50%	1	1
random forest	SHAP	93%	92%	92%	47%	1	1
random forest	perm P2 QL	+528%	+531%	+527%	45%	1	1
random forest	perm P2 MSE	+453%	+458%	+453%	41%	1	1

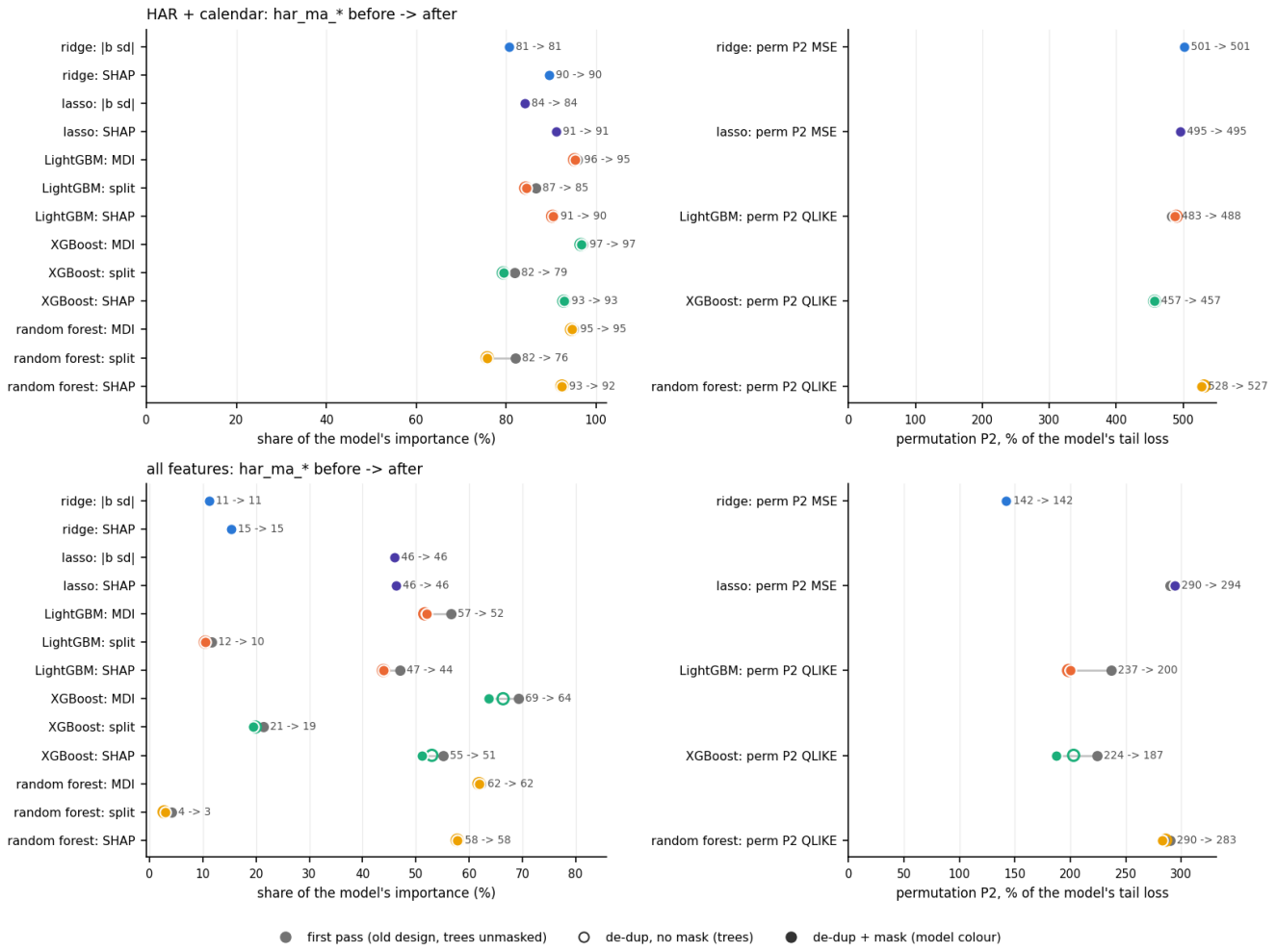


Figure 3: `har_ma_*` before (old design) and after (de-dup + mask; open marker: de-dup, no mask), per model and measure; left: shares, right: permutation P2 as % of the model's tail loss.

- `har_ma_*` in the trees, all_features: LightGBM MDI 57 → 52 %, SHAP 47 → 44 %, perm P2 dQLIKE +237 → +200 % of tail loss (the old `_x_close` copies carried 22 % of the old har_ma column MDI); XGBoost MDI 69 → 64 %, SHAP 55 → 51 %, perm P2 dQLIKE +224 → +187 % of tail loss (the old `_x_close` copies carried 22 % of the old har_ma column MDI); random forest MDI 62 → 62 %, SHAP 58 → 58 %, perm P2 dQLIKE +290 → +283 % of tail loss (the old `_x_close` copies carried 54 % of the old har_ma column MDI).
- The draw-noise yardstick: the ridge fits are the same problem in both runs, so its `har_ma_*` change is the Monte Carlo of fresh permutation draws alone: perm P2 dMSE moves by at most 0.8 points of % tail loss over the 2 buckets (abs(beta x sd), SHAP and drop-column unchanged).
- Spearman old vs new (series level, common units, every bucket x measure): trees median 0.86 (min 0.68), linear median 0.99 (min 0.87); trees old vs de-dup no mask 0.87 (min 0.57), no mask vs mask 0.88 (min 0.66).
- Spearman old vs new (cluster level, common units, every bucket x measure): trees median 0.96 (min 0.59), linear median 1.00 (min 0.77); trees old vs de-dup no mask 0.97 (min 0.56), no mask vs mask 0.98 (min 0.62).
- Spearman old vs new (column level, common units, every bucket x measure): trees median 0.94 (min 0.38), linear median 1.00 (min 0.79); trees old vs de-dup no mask 0.98 (min 0.42), no mask vs mask 0.95 (min 0.40).
- Series leader changed in 4 of 74 bucket x model x measure rows: all_features ridge perm P1 QL `sumret4_ewstock` → `sumautocov`; all_features ridge perm P2 QL `sumautocov` → `voldemand_spx_open_only`; all_features lasso perm P2 QL `har_ma` → `effspread_vwstock`; all_features random forest split `har_ma` → `voldemand_spx_open_and_close`.
- Clusters baseline: matched 19, removed (session-edge only) 6.
- Clusters all_features: matched 230, removed (session-edge only) 6.

- Column level, the dilution the copies caused (permutation P2 dQLIKE x 1e-3 of `har_ma_1` / `har_ma_5` alone, old → new; old = its `_x_close` copy left intact): all features LightGBM 33.4 → 47.4 / 30.7 → 41.0; all features XGBoost 36.0 → 46.8 / 25.9 → 38.3; all features random forest 18.6 → 78.7 / 18.4 → 61.9.
- ridge: perm P2 MSE: `sumbipow_vwstock` 21→18, `sumvolume` 17→19
- lasso: perm P2 MSE: `sumbipow_vwstock` 24→20, `sumautocov` 20→23
- LightGBM: MDI: `sumret` 2→3, `sumabsret` 3→2; split: `sumabsret` 18→9, `stocktwits_sentcount` 13→17; SHAP: `voldemand_spx_open_and_close` 28→20, `sumret3_ewstock` 18→22; perm P2 QL: `sumautocov` 9→40, `buyturnover_ewstock` 12→43; perm P2 MSE: `sumabsret` 33→4, `fomc_until_inv` 36→20
- XGBoost: MDI: `sumbipow` 20→15, `sumret3_ewstock` 16→19; split: `sumabsret` 8→5, `sumret3` 7→9; SHAP: `sumbipow` 17→14, `turnover_vwstock` 14→16; perm P2 QL: `stocktwits_sentcount` 28→14, `sellturnover_ewstock` 31→18; perm P2 MSE: `vix3m` 30→14, `voldemand_all_open_only` 18→31
- random forest: MDI: `stocktwits_sentcount` 17→20, `sumret3_ewstock` 19→16; split: `sumret2_ewstock` 36→12, `vvix` 39→20; SHAP: `buyturnover_ewstock` 15→18, `sumret3_ewstock` 18→15; perm P2 QL: `sumret3_ewstock` 10→29, `ofi_ewstock` 28→9; perm P2 MSE: `ofi_vwstock` 13→39, `vvix` 10→33

Appendix: HAR + calendar design

Table 16: HAR + calendar, series level: top three per model and measure.

model	measure	top three
ridge	$\lvert\beta \cdot sd\rvert$	`har_ma_*` 81%; `calendar` 19%
ridge	perm. P1, ΔMSE	`har_ma_*` +287%; `calendar` +8%
ridge	perm. P2, ΔMSE	`har_ma_*` +501%; `calendar` +9%
ridge	perm. P2, ΔQLIKE	`har_ma_*` +544%; `calendar` +9%
ridge	drop-column, ΔMSE	`har_ma_*` +247%; `calendar` +5%
ridge	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 90%; `calendar` 10%
lasso	$\lvert\beta \cdot sd\rvert$	`har_ma_*` 84%; `calendar` 16%
lasso	perm. P1, ΔMSE	`har_ma_*` +284%; `calendar` +7%
lasso	perm. P2, ΔMSE	`har_ma_*` +495%; `calendar` +7%
lasso	perm. P2, ΔQLIKE	`har_ma_*` +544%; `calendar` +9%
lasso	drop-column, ΔMSE	`har_ma_*` +244%; `calendar` +4%
lasso	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 91%; `calendar` 9%
LightGBM	MDI / gain	`har_ma_*` 95%; `calendar` 5%
LightGBM	split count	`har_ma_*` 85%; `calendar` 15%
LightGBM	perm. P1, ΔQLIKE	`har_ma_*` +224%; `calendar` +8%
LightGBM	perm. P2, ΔQLIKE	`har_ma_*` +488%; `calendar` +8%
LightGBM	perm. P2, ΔMSE	`har_ma_*` +430%; `calendar` +7%
LightGBM	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 90%; `calendar` 10%
XGBoost	MDI / gain	`har_ma_*` 97%; `calendar` 3%
XGBoost	split count	`har_ma_*` 79%; `calendar` 21%
XGBoost	perm. P1, ΔQLIKE	`har_ma_*` +218%; `calendar` +6%
XGBoost	perm. P2, ΔQLIKE	`har_ma_*` +457%; `calendar` +6%
XGBoost	perm. P2, ΔMSE	`har_ma_*` +444%; `calendar` +4%
XGBoost	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 93%; `calendar` 7%
random forest	MDI / gain	`har_ma_*` 95%; `calendar` 5%
random forest	split count	`har_ma_*` 76%; `calendar` 24%
random forest	perm. P1, ΔQLIKE	`har_ma_*` +235%; `calendar` +6%
random forest	perm. P2, ΔQLIKE	`har_ma_*` +527%; `calendar` +6%
random forest	perm. P2, ΔMSE	`har_ma_*` +453%; `calendar` +5%
random forest	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 92%; `calendar` 8%

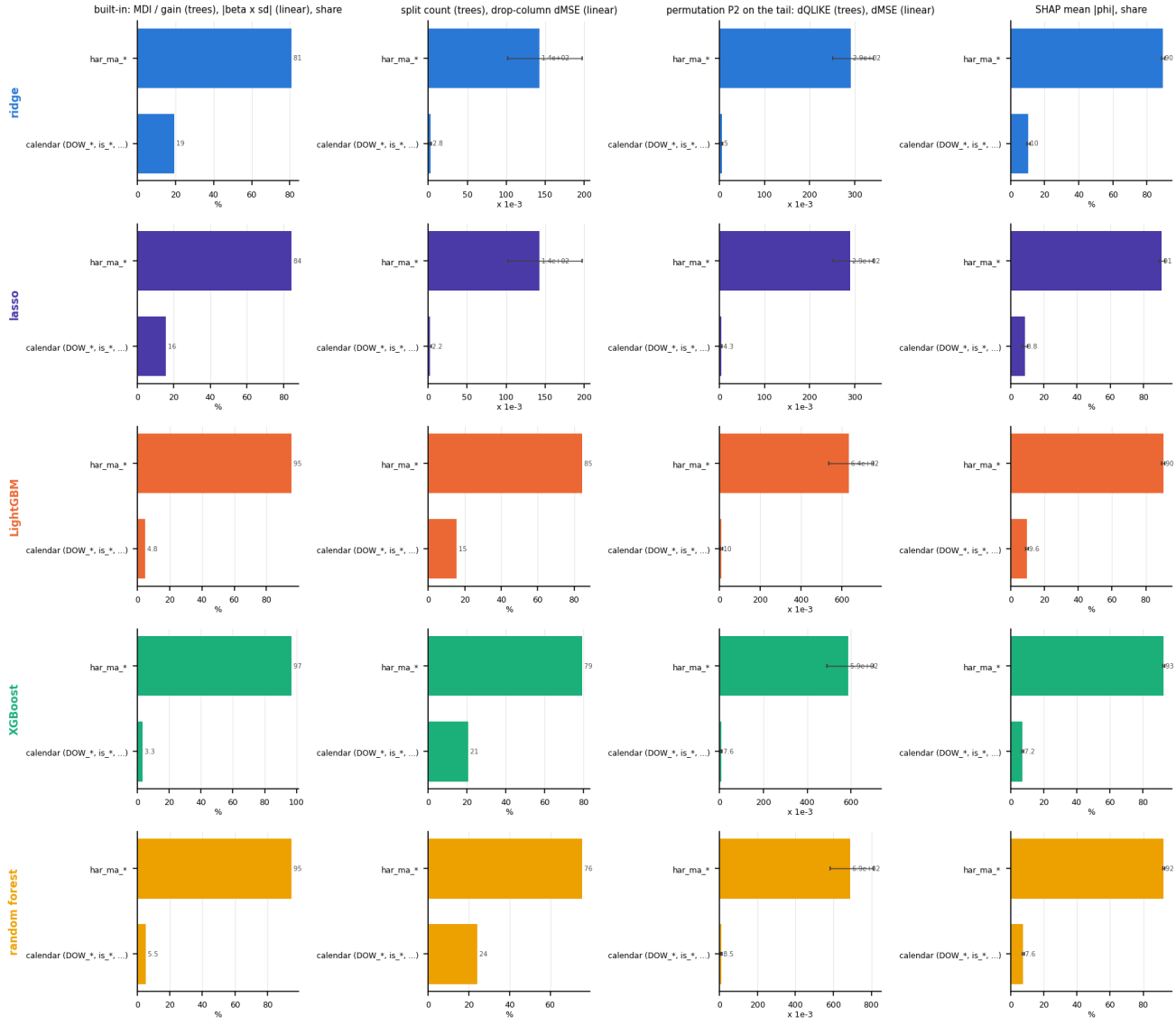


Figure 4: HAR + calendar: ranked bars per measure.

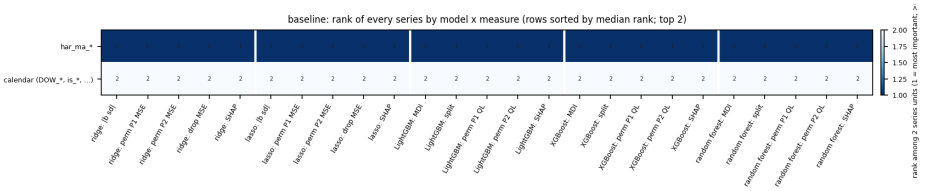


Figure 5: HAR + calendar: ranks across measures.

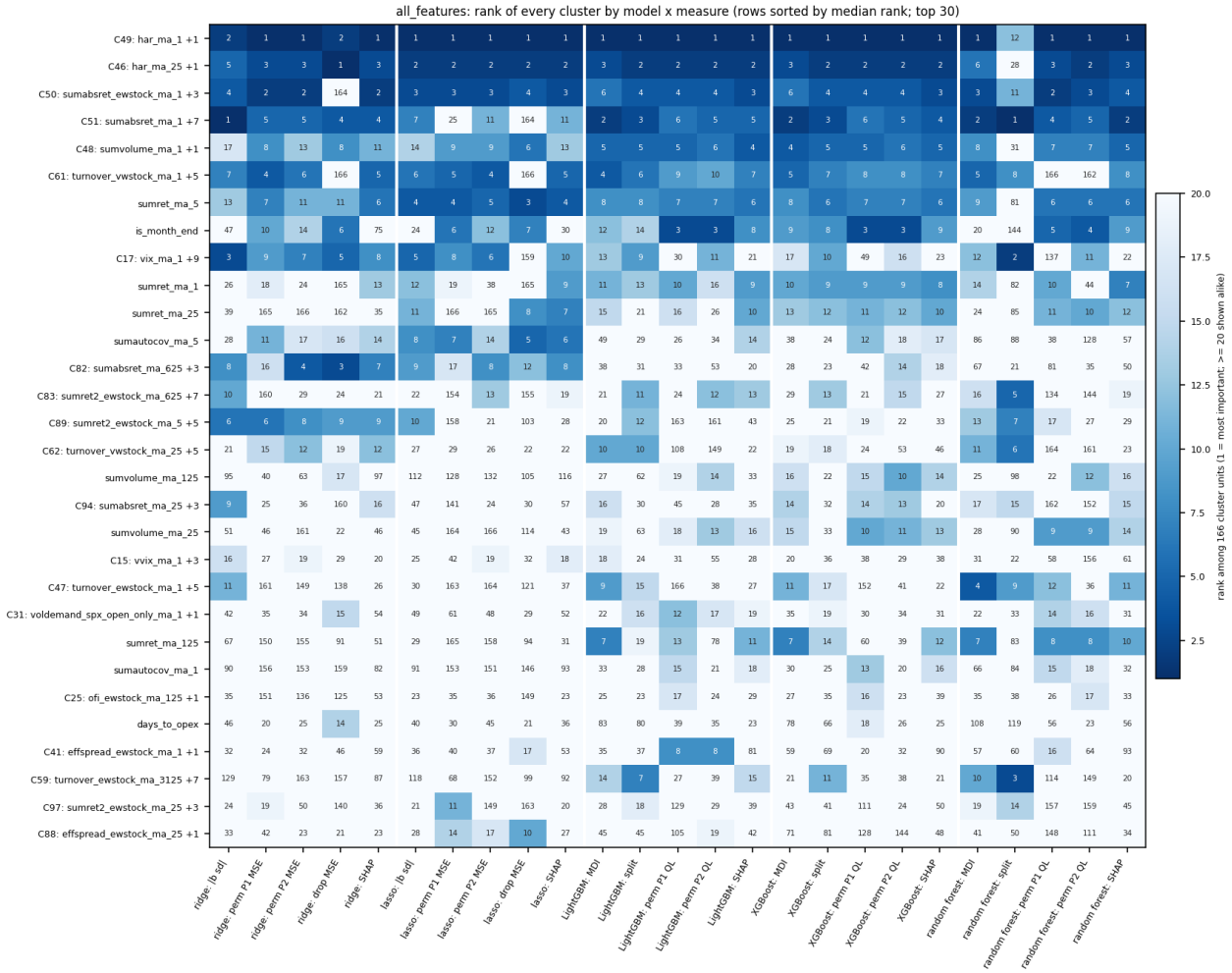


Figure 6: All features: ranks at the cluster level.